MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. The roots of $(x+1)(x-4)$ are
   - A. 1 and 4
   - B. -1 and -4
   - C. -1 and 4
   - D. 1 and -4
2. The discriminant of x^2-2x+3 equals
   - A. -8
   - B. 8
   - C. 16
   - D. -2
3. The vertex of $(x-2)^2+5$ is
   - A. $(2;-5)$
   - B. $(-2;5)$
   - C. $(5;2)$
   - D. $(2;5)$
4. The sum of the roots of $2x^2-6x+1$ equals
   - A. 6
   - B. 3
   - C. -3
   - D. $\frac{1}{2}$
5. $x^2-4<0$ is equivalent to
   - A. $x>2$
   - B. $x<-2$
   - C. $x\in]-2;2[$
   - D. $x\in[-2;2]$

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. x^2+6x+9 has as its factorisation
   - A. $(x-3)^2$
   - B. $(x+9)(x+1)$
   - C. $(x+6)^2$
   - D. $(x+3)^2$
2. The product of the roots of $3x^2+x-2$ equals
   - A. $-\frac{2}{3}$

B. $-\frac{1}{3}$

C. $\frac{2}{3}$

D. -2

3. $-x^2 + 4x$ reaches its maximum on $[0; 1]$ at

A. 2

B. 0

C. 1

D. 4

4. A double root requires

A. $a = 0$

B. $\Delta = 0$

C. $b = 0$

D. $c = 0$

5. The polynomials of degree 2 with roots 0 and 2 are

A. $a(x + 2)$

B. $a(x - 2)^2$

C. $x(x - 2)$ only

D. $ax(x - 2)$, $a \neq 0$

True or false? Justify.

1. A quadratic function always has two real roots.
2. A quadratic with two distinct roots is negative between them.
3. For every real number x, $(x - 2)^2 \geq 0$.

Self-assessment

I justify the choice of form; I distinguish roots from vertex; I finish an inequality with a set; I respect the domain of the model.
For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.